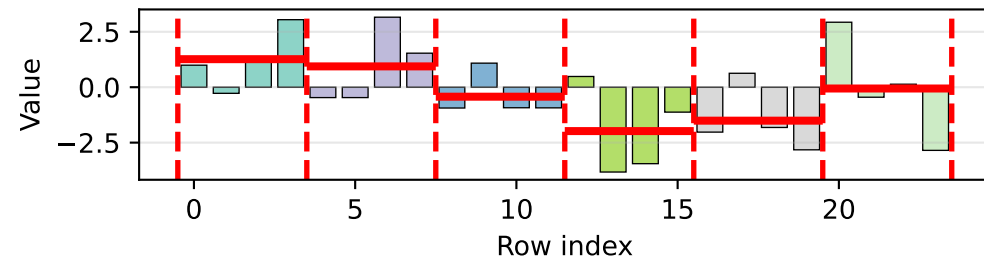
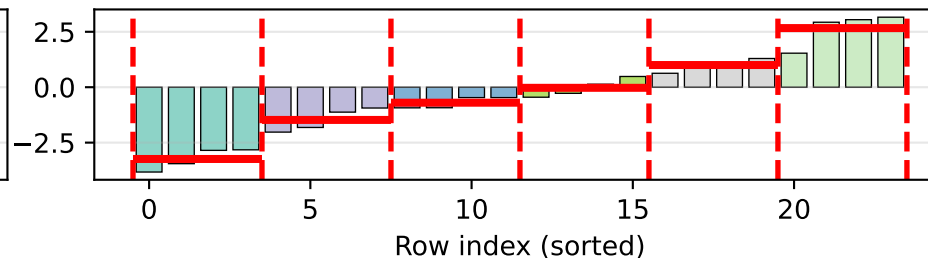


PCA Sort Intuition: Sorting Makes Block Averages More Accurate

Unsorted: Block Mean $\neq$ Row Values
MSE = 2.24



Sorted: Block Mean $\approx$ Row Values
MSE = 0.18 (92% reduction)



WHY SORTING HELPS:

Block-mean replaces each row with its block's average.

If rows in a block are similar, the average is close to each row
→ small error

If rows are different, the average is far from each row
→ large error

SORTING groups similar rows into the same block.

PCA sort is BETTER than value sort because it sorts along the direction where rows vary MOST.

Value sort: $\text{mean}(\text{row}) = \sum x_i / 16$
→ equal weight per dimension

PCA sort: $\text{PC1} \cdot \text{row} = \sum w_i x_i$
→ learned weights from data
→ high-variance dims count more